HARSH RAI

Bina, Madhya Pradesh
+91-8435546284 harshrai1217@gmail.com LinkedIn Github Portfolio

EDUCATION

Indian Institute of Information Technology Jabalpur **Aug 2023 – Aug 2027**
Bachelor of Technology Computer Science and Engineering *Jabalpur, India*

TECHNICAL SKILLS / COURSEWORK

Languages: C, C++, Python, Html, CSS, JavaScript, SQL, NoSQL
Tech./Frameworks: Git, GitHub, ReactJS, Redux, NodeJS, ExpressJS, FastApi, Mongodb, Bootstrap, Tailwind CSS
AI/GenAI: LLM Integration, Prompt Engineering, Agentic AI Basics, n8n Workflow Automation
Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Computer Architecture, Software Engineering, Artificial Intelligence
Core Competencies: Communication, Teamwork, Problem Solving, Analytical Thinking

EXPERIENCE

Bosch Global Software Technologies **Bengaluru, India**
Software Engineer Intern *May 2026 - Jul 2026*

- Gained foundational knowledge of **ERP systems, SAP BTP, and SAP Fiori**, along with Supply Chain Management and enterprise IT workflows, before project development.
- Built an **AI-powered Knowledge Agent** with **text and voice interfaces** across **6 knowledge domains**, applying Generative AI and LLM integration to capture and structure employee expertise.
- Worked full-stack across **React.js, Python FastAPI, and SQLite**, building **RESTful APIs** across a **4-table database schema** connecting the frontend, backend, and database layers.
- Worked with **2 LLMs** (via Groq) to balance speed and accuracy across each stage of the extraction pipeline.
- Strengthened full-stack development and AI engineering capabilities through hands-on enterprise experience.

PROJECTS

Helix AI | React, FastAPI, SQLite, Groq LLM **Jul 2026**

- Built an AI-powered knowledge agent during my internship at Bosch, with text and voice interfaces to capture employee expertise through conversational interviews.
- Engineered LLM-driven pipelines to convert free-form employee answers into structured, confidence-scored knowledge chunks.
- Designed a normalized SQLite database and REST APIs with FastAPI to store and query the knowledge base.
- Built a retrieval-augmented query system (Ask Alex) that fetches relevant knowledge chunks and generates cited answers via LLM.

Apna-VideoCall | MongoDB, Express.js, React, Node.js **Aug 2025**

- Built a real-time video conferencing platform using the MERN stack, integrating WebRTC for peer-to-peer audio/video streaming and Socket.IO for real-time signaling and messaging.
- Implemented core meeting features including instant chat, dynamic room creation and management, screen sharing, and live in-app notifications.
- Designed a responsive React.js UI with adaptive video grid layouts and participant controls, and used Redux to manage state across sessions, chat, and media streams.
- Built RESTful APIs with Node.js and Express.js to handle user and session data, using MongoDB for storage, and deployed the platform on Render.

WanderStay | MongoDB, Express.js, React, Node.js **Mar 2025**

- Built a full-featured travel booking platform using the MERN stack, enabling users to explore listings, reserve accommodations, and manage bookings end-to-end.
- Delivered a responsive, mobile-friendly UI with React.js, React Router, and Context API for client-side state, ensuring smooth navigation and reusable components across pages.
- Integrated secure authentication and authorization with JWT and bcrypt, handling signup, login, and protected routes for user sessions.
- Built RESTful APIs with Node.js, Express.js, and MongoDB Atlas to manage users, packages, and booking workflows, with image uploads via Multer and Cloudinary, deployed on Render.

Achievements

- **Oracle OCI AI Foundations Certification** — AI, ML, Deep Learning, Generative AI on Oracle Cloud.
- **Oracle Agentic AI Foundations Certification** —Core concepts of agentic AI systems.
- **Google AI Professional Certificate** — Applying AI tools to real-world business problems.
- **Google AI Essentials** — Foundations of Generative AI concepts and applications.
- **Google Prompting Essentials** — Practical techniques for effective Prompt Engineering.
- **300+ Coding Problems Solved** (CodeChef, Codeforces, LeetCode) — Sharpened problem-solving skills.